Computer Organization and Architecture (CSL251)
LAB-10
2025-26 W

Pipelined Processor Demonstration using RISC-V Assembly

Pipelined Processor Demonstration

- 5-Stage Pipeline.

Stages:

- Instruction Fetch (IF)
- Instruction Decode (ID)
- Execute (EX)
- Memory Access (MEM)
- Write Back (WB)

Pipelined Processor Demonstration

Components used:

- **Instruction Cache (ICache)-** Stores machine instructions.
- **Register File (RF)-** Stores register values.
- **Program Counter (PC)-** Points to the current instruction.
- **ALU-** Performs arithmetic and logical operations.
- **Control Logic-** Decodes the instruction.

Pipelined Processor Demonstration

- Example contains three arithmetic/logic instructions

Instructions used:

- ADD R1, R2, R3
- SUB R4, R5, R6
- AND R7, R8, R9
- HLT

Pipelined Processor Demonstration

- Instruction Format

opcode[15:12]	REG1[11:8]	REG2[7:4]	REG3[3:0]
4 bits	4 bits	4 bits	4 bits

Field	Description
Opcode	Operation type
REG1	Destination register
REG2	Source register
REG3	Source register

Pipelined Processor Demonstration

- Instruction Cache

Instruction	Machine Code	Opcode meaning
ADD R1,R2,R3	0x0123	0
SUB R4,R5,R6	0x1456	1
AND R7,R8,R9	0x4789	4
HLT	0xF000	15

Pipelined Processor Demonstration

Register File

- The processor contains 16 registers.
- Each register is 8 bits
- Registers are named R0 – R15
- R0 always stores value 0.
- Other registers are initialized to 0.

Register	Value
R2	2
R3	3
R5	7
R6	4
R8	6
R9	5

Pipelined Processor Demonstration

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Pipelined Processor Demonstration

Program Counter (PC)

PC: .word 0

- PC = 0 → first instruction
- PC = 2 → next instruction
- PC = 4 → next instruction
- PC = 6 → HLT

Each instruction is 2 bytes, so:

PC = PC + 2

Pipelined Processor Demonstration

- **Pipeline stages:**

$$IF \rightarrow ID \rightarrow EX \rightarrow MEM \rightarrow WB$$

Cycle	IF	ID	EX	MEM	WB
1	ADD				
2	SUB	ADD			
3	AND	SUB	ADD		
4	HLT	AND	SUB	ADD	
5		HLT	AND	SUB	ADD

Pipelined Processor Demonstration

- **After executing all instructions:**

ADD R1,R2,R3

R1 = 2 + 3 = 5

SUB R4,R5,R6

R4 = 7 - 4 = 3

AND R7,R8,R9

R7 = 6 & 5 = 4